

2008 Ford Edge SE

2008 DRIVELINE/AXLES Front Drive Halfshafts - Edge & MKX

2008 DRIVELINE/AXLES

Front Drive Halfshafts - Edge & MKX

SPECIFICATIONS

GENERAL SPECIFICATIONS

Item	Specification
Left front halfshaft assembled length	671.8 mm (26.44 in)
Right front halfshaft assembled length (all wheel drive [AWD])	632.4 mm (24.89 in)
Right front halfshaft assembled length (front wheel drive [FWD])	605.4 mm (23.83 in)

LUBRICANTS & CAPACITIES

Lubricant	Specification	Capacity
Front Halfshaft Inboard Constant Velocity (CV) Joint		
Constant Velocity Joint Grease (from 3A331 kit)	-	295 g (10.41 oz)
Front Halfshaft Outboard CV Joint		
Constant Velocity Joint Grease (from 3A331 kit)	-	130 g (4.58 oz)

TORQUE SPECIFICATIONS

Description	Nm	lb-ft
Ball joint pinch nut	55	41
Catalytic converter support bracket bolts	20	15
Front stabilizer bar link nut	90	66
Front wheel hub nut	350	258
Intermediate shaft bearing support bolts (AWD)	40	30
Intermediate shaft bearing support bolts (FWD)	40	30

DESCRIPTION AND OPERATION

FRONT DRIVE HALFSHAFTS

The front drive halfshafts consist of the following:

- Inboard and outboard constant velocity (CV) joints connect to a splined shaft. Driveshaft bearing retainer circlips retain the CV joints to the splined shaft.
- A front axle wheel hub nut that secures the splined outboard CV joint to the wheel hub.
- Lubed-for-life CV joints use special CV joint grease requiring no periodic lubrication.
- A boot repair kit, 3A331, which includes the boots, clamps and grease is available.

On the LH side, the inboard CV joint is retained in the differential side gear with a 33 mm (1.29 in) retainer circlip. This circlip is oval in shape when viewed at the end. Install a new circlip every time the halfshaft is disconnected. On the RH side, a 30 mm (1.181 in) driveshaft bearing retainer circlip retains the splined inboard CV joint to the intermediate shaft. This circlip is round when view at the end. Install a new circlip every time you disconnect the halfshaft from the intermediate shaft.

The halfshafts are splined on the outboard stub shaft to drive the wheel hubs. They are retained in the wheel hubs by special wheel hub nuts which also control the wheel bearing preload. The LH halfshaft is splined on the inboard stub shaft and is retained in the differential side gear in the transaxle by a circlip. The RH halfshaft has internal splines which are driven by the intermediate shaft. The intermediate shaft has a circlip on the outboard end and is retained inside the inboard stub shaft. The circlips must be installed new whenever they are removed. The intermediate shaft is retained in the transaxle differential side gear by bolts that go through the intermediate shaft support bearing. The intermediate shaft also goes through the power transfer unit (PTU) on all wheel drive (AWD) vehicles. The outer seal of the PTU must be installed new whenever the intermediate shaft is removed.

Halfshaft Handling

CAUTION: Never pick up or hold the halfshaft only by the inboard or outboard constant velocity (CV) joint or damage to the component may occur.

Handle all halfshaft components carefully during removal and installation and during various component disassembly and assembly procedures. Do not overangle the CV joints. Damage will occur to an assembled inboard CV joint if it is overplunged outward from the joint housing. Never use the halfshaft assembly as a lever to position other components. Always support the free end of the halfshaft. Do not allow the boots to contact sharp edges or hot exhaust components. Handle the halfshaft only by the interconnecting shaft to avoid pull-apart and potential damage to the CV joints. Excessive pulling force on the interconnecting shaft between joints of the halfshaft will result in internal joint damage. Axial loads used in assisting removal must be applied through the inboard joint housing only. Do not drop assembled halfshafts. The impact will cut the boots from the inside without evidence of external damage.

DIAGNOSTIC TESTS

FRONT DRIVE HALFSHAFTS

Refer to DRIVELINE SYSTEM - GENERAL INFORMATION article.

REMOVAL AND INSTALLATION

HALFSHAFT - RH

Special Tools

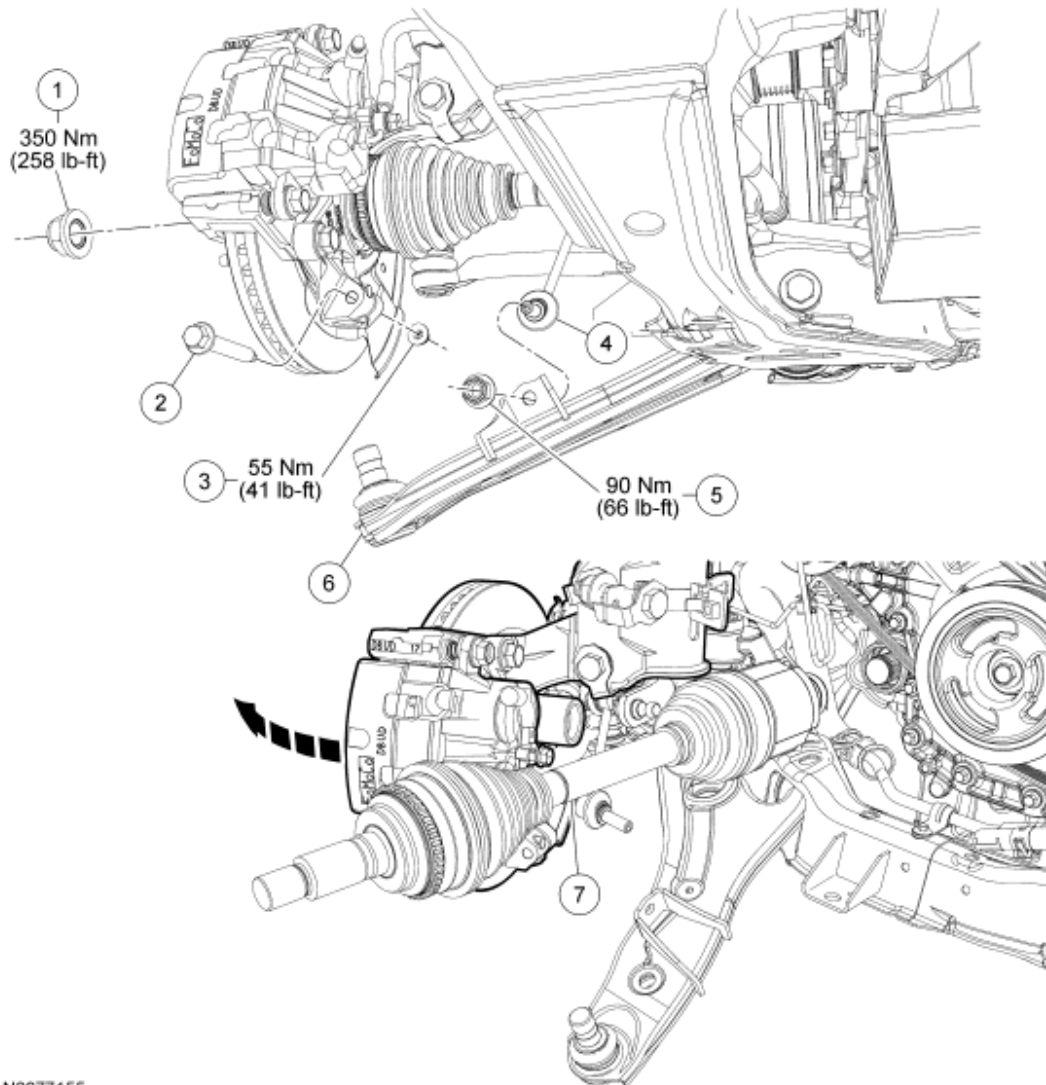
Illustration	Tool Name	Tool Number
	Remover, Front Wheel Hub	205-D070 (D93P-1175-B) or equivalent

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ST2272-A



N0077155

Fig. 1: Identifying Halfshaft With Torque Specifications - RH
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	3N404	Wheel hub nut
2	W712481	Ball joint pinch bolt
3	W520213	Ball joint pinch nut
4	3B438	Stabilizer bar link

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5	W520214	Stabilizer bar link nut
6	3A423	Lower arm
7	3A427	Halfshaft assembly

REMOVAL

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING & LIFTING** article.
2. Remove the front tire and wheel. For additional information, refer to **WHEELS & TIRES** article.

NOTE: Apply the brake to keep the halfshaft from rotating.

3. Remove and discard the front wheel hub nut.

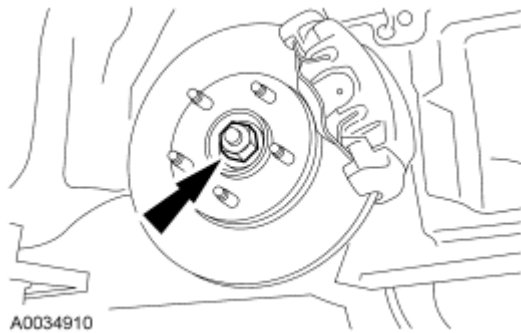


Fig. 2: Locating Front Wheel Hub Nut
Courtesy of FORD MOTOR CO.

4. Remove the stabilizer bar link nut and position the link aside.

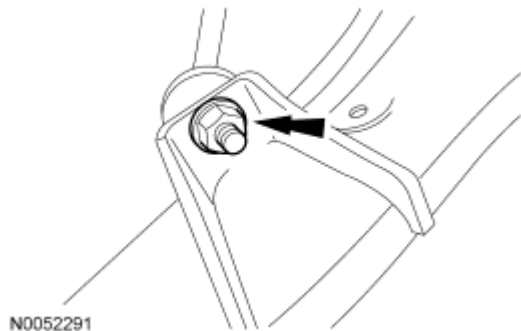
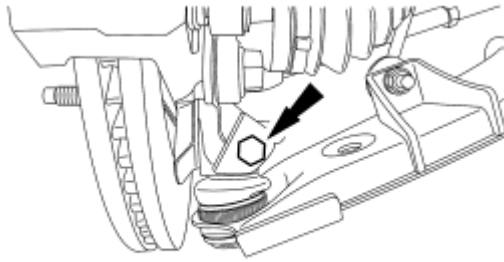


Fig. 3: Locating Stabilizer Bar Link Nut
Courtesy of FORD MOTOR CO.

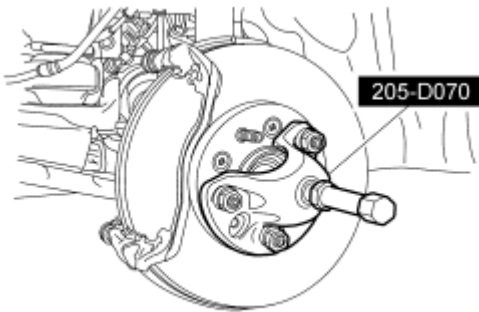
5. Remove the ball joint pinch bolt from the knuckle and separate the lower control arm.



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Fig. 4: Locating Ball Joint Pinch Bolt
Courtesy of FORD MOTOR CO.

6. Using the special tool, separate the halfshaft from the wheel hub.

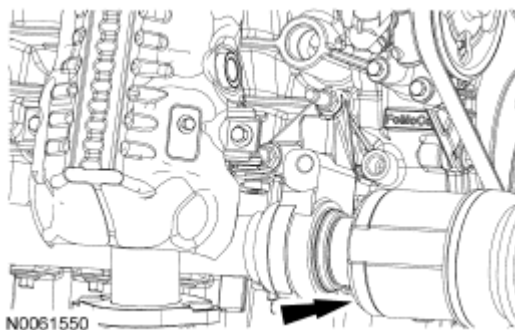


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Fig. 5: Separating Halfshaft From Wheel Hub Using Special Tool (205-D070)
Courtesy of FORD MOTOR CO.

NOTE: Support the knuckle with a suitable jackstand.

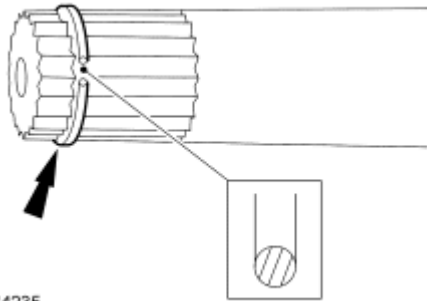
7. Pull the knuckle outboard and rotate it toward the rear of the vehicle.
8. Use a brass drift to strike the right side halfshaft in the indicated area and separate the RH halfshaft from the intermediate shaft.



N0061550

Fig. 6: Identifying Right Side Halfshaft
Courtesy of FORD MOTOR CO.

9. Remove and discard the circlip from the intermediate shaft.

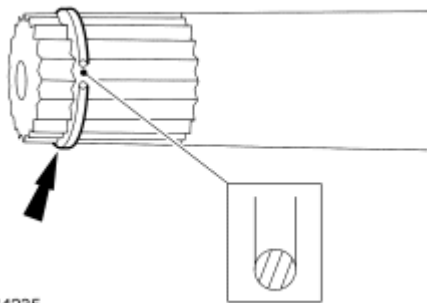


N0064235

Fig. 7: Identifying Circlip From Intermediate Shaft
Courtesy of FORD MOTOR CO.

INSTALLATION

1. Install a new 30 mm (1.181 in) intermediate shaft circlip.

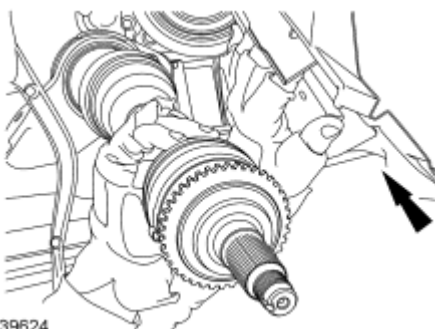


N0064235

Fig. 8: Identifying Circlip From Intermediate Shaft
Courtesy of FORD MOTOR CO.

NOTE: Pull the right side inboard joint outward to make sure the circlip is locked.

2. Align the splines on the right side shaft with the intermediate shaft and push the stub shaft in until the circlip locks the shafts together.

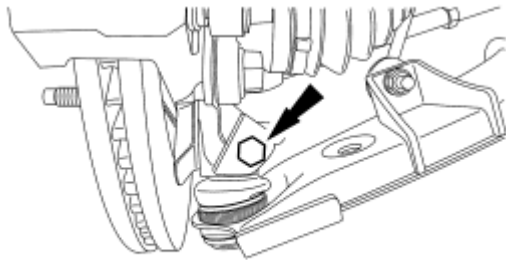


N0039624

Fig. 9: Locating Pushpins

Courtesy of FORD MOTOR CO.

3. Insert the halfshaft into the wheel hub.
4. Rotate the knuckle into position.
5. Install the ball joint in the knuckle and install the pinch bolt and nut.
 - Tighten to 55 Nm (41 lb-ft).

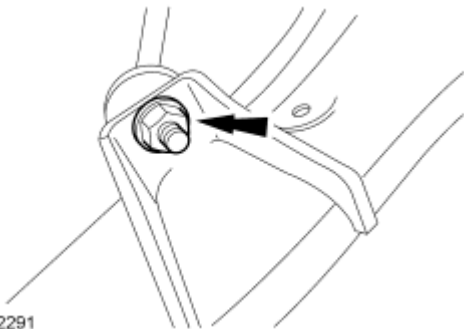


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Fig. 10: Locating Ball Joint Pinch Bolt

Courtesy of FORD MOTOR CO.

6. Position the stabilizer bar link and install the link nut.
 - Tighten to 90 Nm (66 lb-ft).

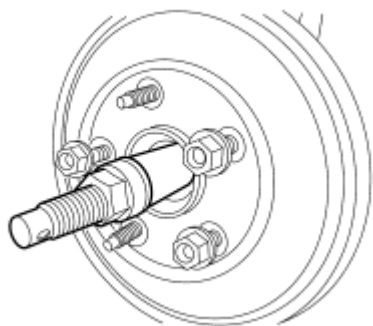


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Fig. 11: Locating Stabilizer Bar Link Nut

Courtesy of FORD MOTOR CO.

7. Using a suitable tool, install the halfshaft in the wheel hub.



N0059422

Fig. 12: Locating Halfshaft On Wheel Hub
Courtesy of FORD MOTOR CO.

NOTE: Do not tighten the front wheel hub nut with the vehicle on the ground. The nut must be tightened to specification before the vehicle is lowered onto the wheels. Wheel bearing damage may occur if the wheel bearing is loaded with the weight of the vehicle applied.

NOTE: Apply the brake to keep the halfshaft from rotating.

8. Install a new front wheel hub nut.
 - Tighten to 350 Nm (258 lb-ft).
9. Install the front tire and wheel. For additional information, refer to **WHEELS & TIRES** article.

HALFSHAFT - LH

Special Tools

Illustration	Tool Name	Tool Number
 ST2272-A	Remover, Front Wheel Hub	205-D070 (D93P-1175-B) or equivalent
 ST1582-A	Remover, Halfshaft	205-241
 ST2038-A	Remover, Halfshaft (Plate)	205-290

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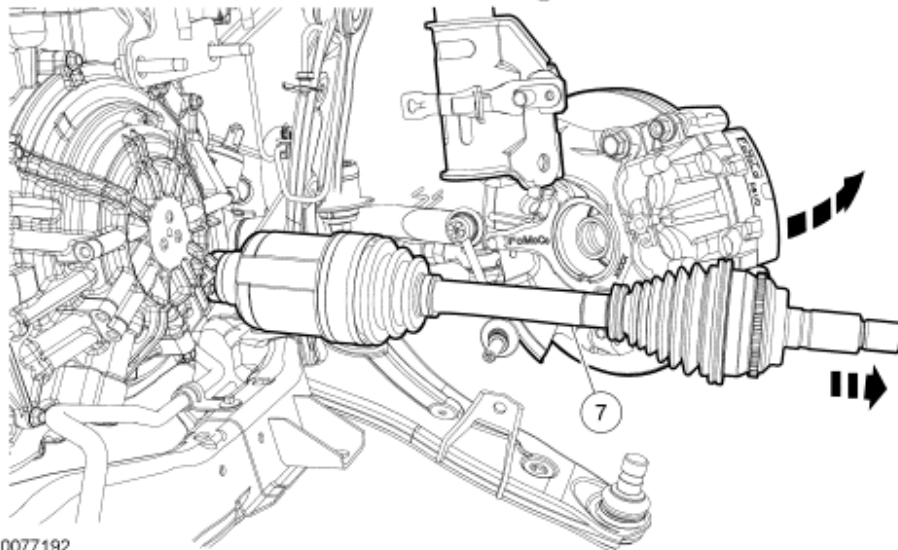
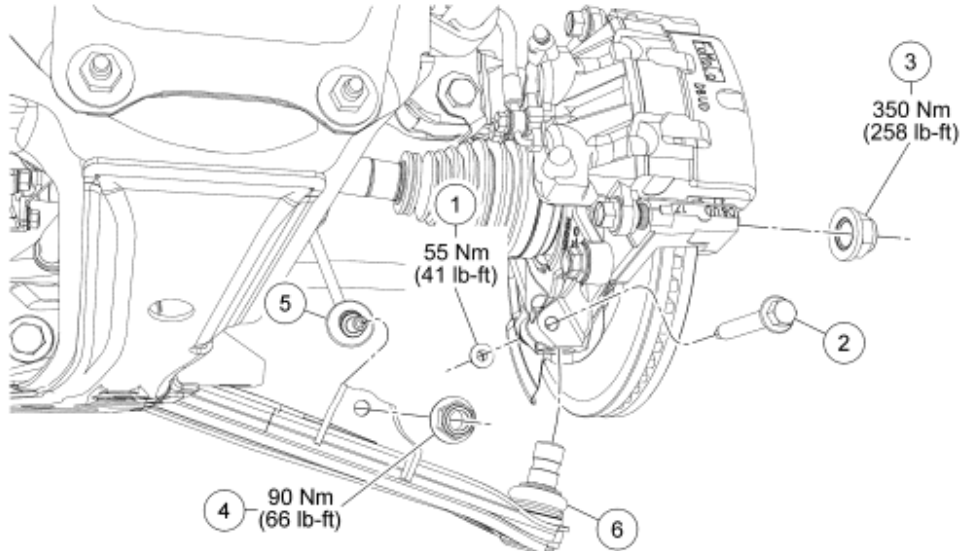
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ST1185-A

Slide Hammer

100-001 (T50T-100-A)



N0077192

Fig. 13: Identifying Halfshaft With Torque Specifications - LH
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W520213	Lower ball joint pinch nut
2	W712481	Lower ball joint pinch bolt
3	3N404	Wheel end nut and washer assembly
4	W520214	Stabilizer bar link nut

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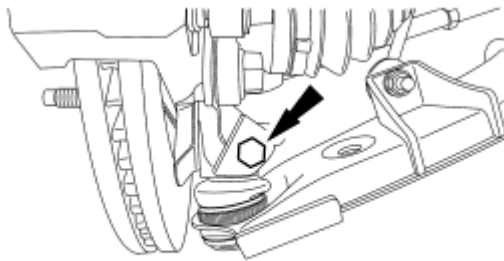
5	3B438	Stabilizer bar link
6	3A423	Lower arm
7	3A428	Halfshaft assembly

REMOVAL

1. With the vehicle in NEUTRAL, position it on a hoist. For additional information, refer to **JACKING & LIFTING** article.
2. Remove the front tire and wheel. For additional information, refer to **WHEELS & TIRES** article.

NOTE: Apply the brake to keep the halfshaft from rotating.

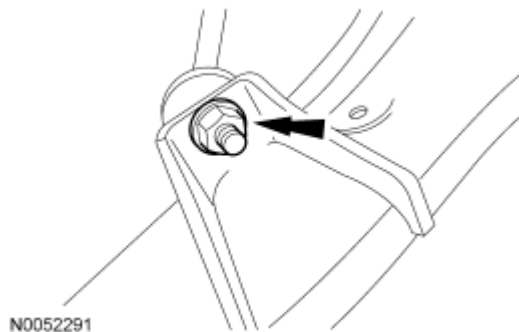
3. Remove and discard the front wheel hub nut.
4. Remove the ball joint pinch bolt from the knuckle and separate the lower control arm.



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Fig. 14: Locating Ball Joint Pinch Bolt
Courtesy of FORD MOTOR CO.

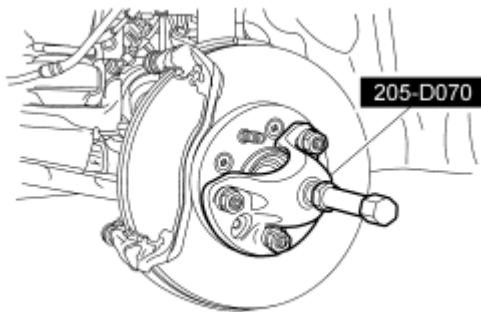
5. Remove the stabilizer bar link nut and position the link aside.



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Fig. 15: Locating Stabilizer Bar Link Nut
Courtesy of FORD MOTOR CO.

6. Using the special tool, separate the halfshaft from the wheel hub.

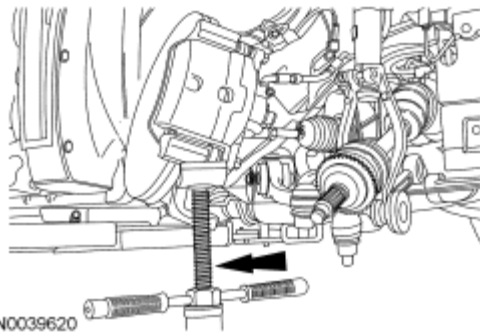


N0040222

Fig. 16: Separating Halfshaft From Wheel Hub Using Special Tool (205-D070)
Courtesy of FORD MOTOR CO.

NOTE: Support the knuckle with a suitable jackstand.

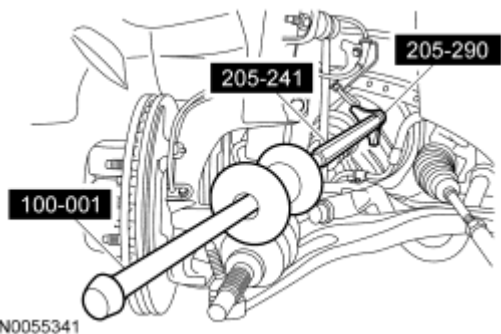
7. Pull the knuckle outboard and rotate it toward the front of the vehicle.



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Fig. 17: Locating Jackstand
Courtesy of FORD MOTOR CO.

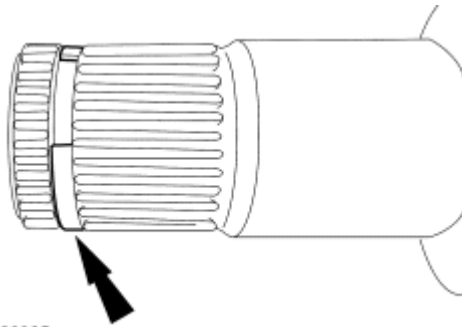
8. Using the special tools, remove the halfshaft from the transmission.



N0055341

Fig. 18: Removing Halfshaft Using Special Tools (100-001), (205-241) & (205-290)
Courtesy of FORD MOTOR CO.

9. Remove and discard the circlip from the stub shaft.

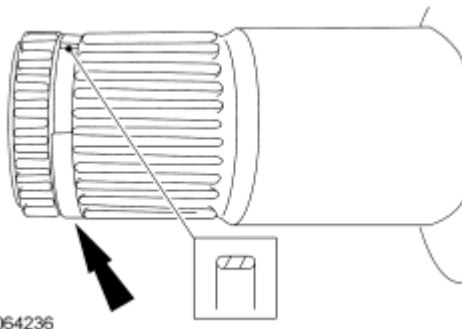


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Fig. 19: Locating Halfshaft Circlip
Courtesy of FORD MOTOR CO.

INSTALLATION

1. Install a new stub shaft circlip.



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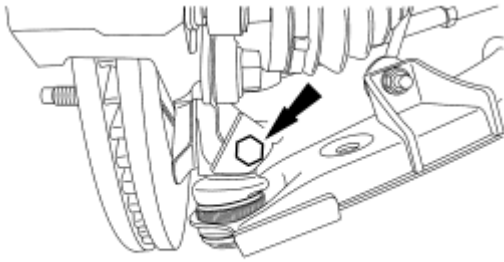
Fig. 20: Installing New Stub Shaft Circlip
Courtesy of FORD MOTOR CO.

2. Insert the halfshaft into the wheel hub.

CAUTION: The sharp edges on the stub shaft splines can slice or puncture the oil seal. Use care when inserting the stub shaft into the transmission to avoid oil seal damage.

NOTE: After insertion, pull the halfshaft inner end to make sure the circlip is locked.

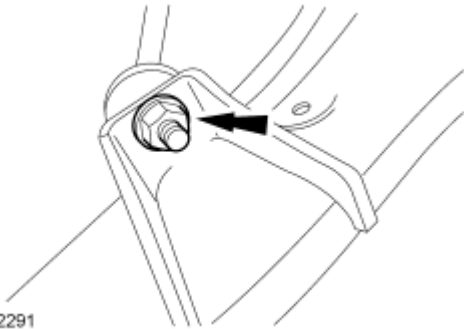
3. Push the stub shaft into the transmission so the circlip locks into the differential side gear.
4. Rotate the knuckle into position.
5. Install the ball joint in the knuckle and install the pinch bolt and nut.
 - Tighten to 55 Nm (41 lb-ft).



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Fig. 21: Locating Ball Joint Pinch Bolt
Courtesy of FORD MOTOR CO.

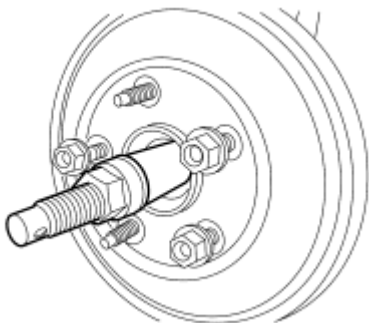
6. Position the stabilizer bar link and install the link nut.
 - Tighten to 90 Nm (66 lb-ft).



N0052291

Fig. 22: Locating Stabilizer Bar Link Nut
Courtesy of FORD MOTOR CO.

7. Using a suitable tool, install the halfshaft in the wheel hub.



N0059422

Fig. 23: Locating Halfshaft On Wheel Hub
Courtesy of FORD MOTOR CO.

CAUTION: Do not tighten the front wheel hub nut with the vehicle on the ground. The nut must be tightened to specification before the vehicle is lowered onto the wheels. Wheel bearing damage will occur if the

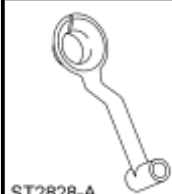
wheel bearing is loaded with the weight of the vehicle applied.

NOTE: Apply the brake to keep the halfshaft from rotating.

- 8. Install a new front wheel hub nut.
 - Tighten to 350 Nm (258 lb-ft).
- 9. Install the front tire and wheel. For additional information, refer to **WHEELS & TIRES** article.

INTERMEDIATE SHAFT - FRONT WHEEL DRIVE (FWD)

Special Tools

Illustration	Tool Name	Tool Number
	Seal Protector, Rear Axle Shaft	205-816

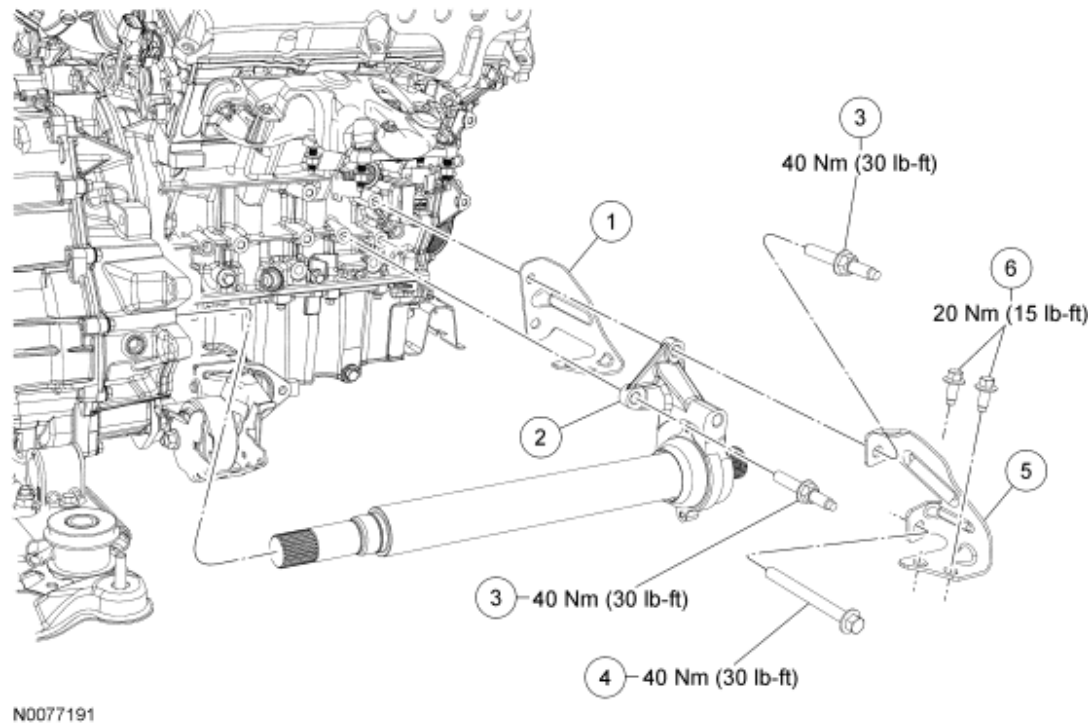


Fig. 24: Exploded View Of Intermediate Shaft - Front Wheel Drive (FWD) With Torque Specifications
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	5G288	Exhaust converter bracket

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2	3K183	Intermediate shaft assembly
3	W500235-S	Bracket-to-block bolts (2 required)
4	W500131	Bracket-to-block bolt
5	5G288	Exhaust converter bracket
6	W500210	Exhaust converter bracket bolts (2 required)

REMOVAL

1. Remove the right halfshaft. For additional information, refer to **Halfshaft - RH**.
2. Remove the 2 RH catalytic converter support bracket bolts.

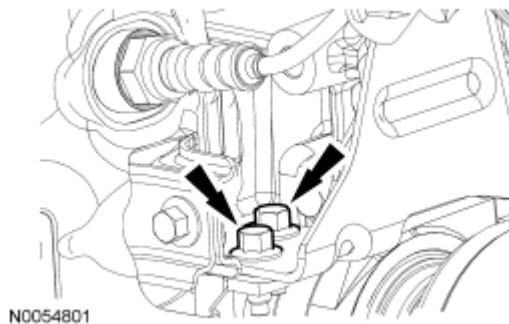


Fig. 25: Identifying RH Catalytic Converter Support Bracket Bolts
Courtesy of FORD MOTOR CO.

3. Remove the bolts from RH catalytic converter support bracket.

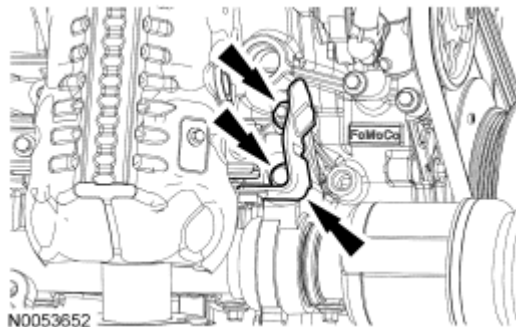
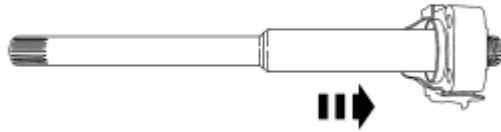


Fig. 26: Identifying RH Catalytic Converter Support Bracket & Bolts
Courtesy of FORD MOTOR CO.

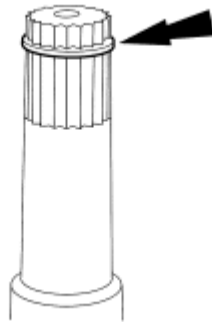
4. Remove the intermediate shaft bearing support bracket-to-block bolt.
5. Carefully remove the intermediate shaft while supporting both ends of the intermediate shaft.



N0060269

Fig. 27: Identifying Intermediate Shaft
Courtesy of FORD MOTOR CO.

6. Remove and discard the circlip from the outboard end of the intermediate shaft.

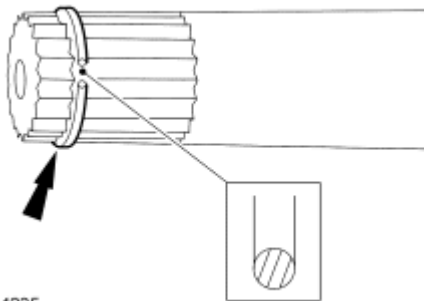


A0000438

Fig. 28: Locating Intermediate Shaft Circlip
Courtesy of FORD MOTOR CO.

INSTALLATION

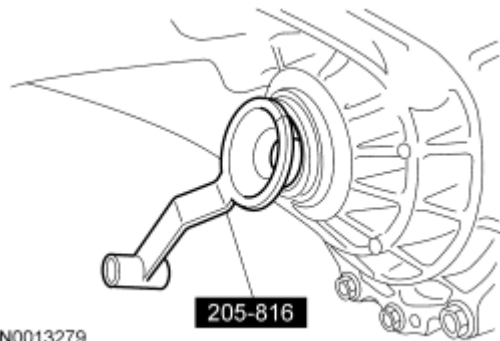
1. Install a new 30 mm (1.181 in) circlip on the outboard end of the intermediate shaft.



N0064235

Fig. 29: Identifying Circlip From Intermediate Shaft
Courtesy of FORD MOTOR CO.

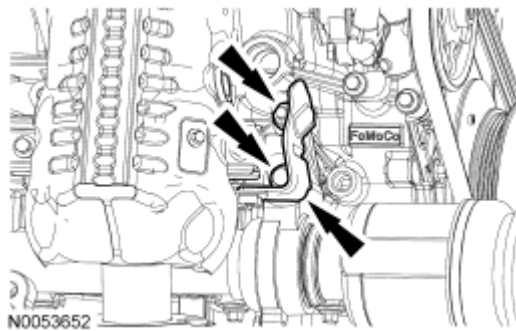
2. Position the special tool over the halfshaft oil seal.



N0013279

Fig. 30: Identifying Special Tool (205-816)
Courtesy of FORD MOTOR CO.

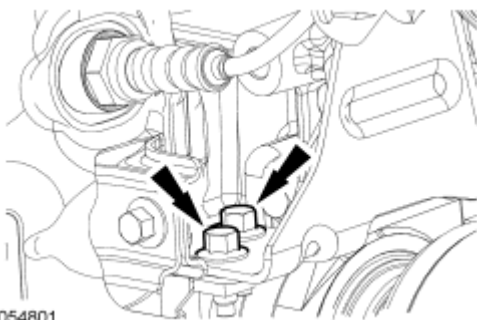
3. Position the intermediate shaft in the transaxle and engage the intermediate shaft splines with the transaxle side gears. Make sure the circlip is locked in the gear.
4. Install the intermediate shaft support bracket-to-block stud bolt.
 - Tighten to 40 Nm (30 lb-ft).
5. Install the RH catalytic converter support brackets and bolts.
 - Tighten to 40 Nm (30 lb-ft).



N0053652

Fig. 31: Identifying RH Catalytic Converter Support Bracket & Bolts
Courtesy of FORD MOTOR CO.

6. Install the 2 RH catalytic converter support bracket bolts.
 - Tighten to 20 Nm (15 lb-ft).



N0054801

Fig. 32: Identifying RH Catalytic Converter Support Bracket Bolts
Courtesy of FORD MOTOR CO.

7. Install the right halfshaft. For additional information, refer to **Halfshaft - RH**.

INTERMEDIATE SHAFT - ALL WHEEL DRIVE (AWD)

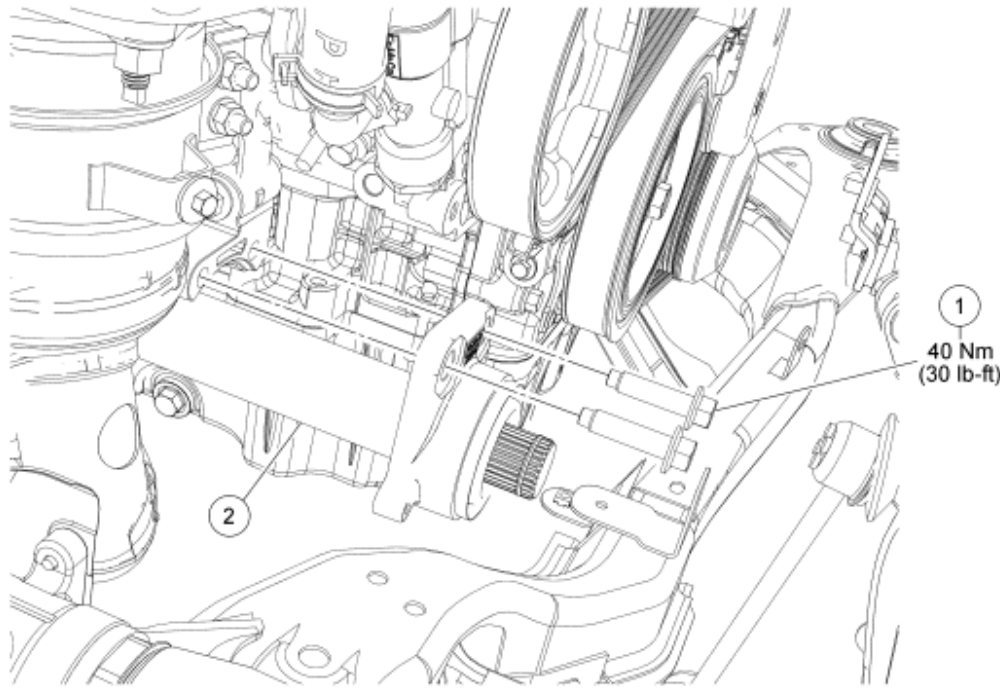


Fig. 33: Exploded View Of Intermediate Shaft - All Wheel Drive (AWD) With Torque Specification
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	W500235-S	Intermediate shaft support bracket bolt (2 required)
2	3K183	Intermediate shaft assembly

REMOVAL

NOTE: The intermediate shaft seal in the Power Transfer Unit (PTU) must be replaced whenever the intermediate shaft is removed. For additional information, refer to **TRANSFER CASE - POWER TRANSFER UNIT (PTU)** article.

1. Remove the right halfshaft assembly. For additional information, refer to **Halfshaft - RH**.
2. Remove the 2 intermediate shaft support bracket bolts and the intermediate shaft.
3. Remove and discard the circlip from the outboard end of the intermediate shaft.

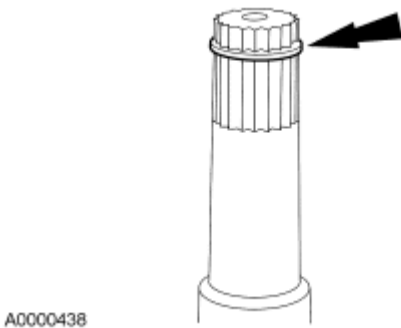


Fig. 34: Locating Intermediate Shaft Circlip
Courtesy of FORD MOTOR CO.

INSTALLATION

- 1. Install a new 30 mm (1.181 in) circlip on the outboard end of the intermediate shaft.

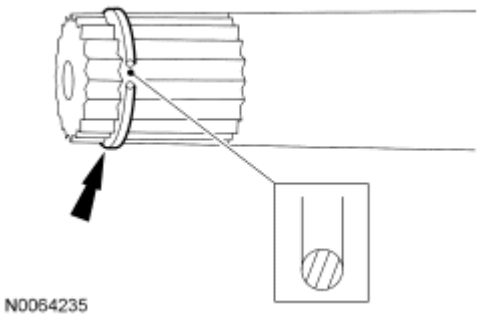


Fig. 35: Identifying Circlip From Intermediate Shaft
Courtesy of FORD MOTOR CO.

- 2. Install a new intermediate shaft seal in the PTU. For additional information, refer to **TRANSFER CASE - POWER TRANSFER UNIT (PTU)** article.
- 3. Position the intermediate shaft in the PTU and engage the intermediate shaft splines with the PTU gears.
- 4. Install the 2 intermediate shaft support bracket bolts.
 - Tighten to 40 Nm (30 lb-ft).
- 5. Install the right halfshaft. For additional information, refer to **Halfshaft - RH.**

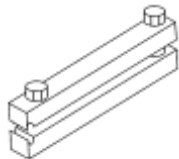
CONSTANT VELOCITY (CV) JOINT BOOT

Special Tools

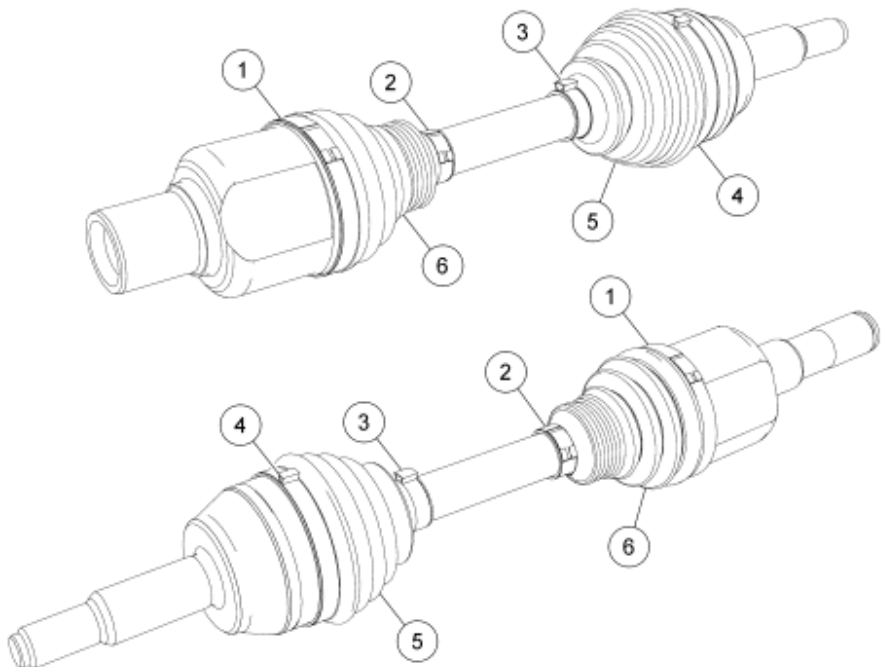
Illustration	Tool Name	Tool Number
	Installer, Constant Velocity Joint Boot Clamp	205-343 (T95P-3514-A)

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ST1301-A



N0061549

Fig. 36: Exploded View Of Constant Velocity (CV) Joint Boot
Courtesy of FORD MOTOR CO.

Item	Part Number	Description
1	-	Low profile large inboard boots inner clamps (serviced in 3A331 kit only)
2	-	Low profile small inboard boots (serviced in 3A331 kit only)
3	-	Intermediate shaft boot clamps (serviced in 3A331 kit only)
4	-	Outboard boot outer clamps (serviced in 3A331 kit only)
5	-	Outboard constant velocity (CV) joint boots (serviced in 3A331 kit only)
6	-	Inboard CV joint boots (serviced in 3A331 kit only)

1. Remove the halfshaft assembly. For additional information, refer to **Halfshaft - RH** and **Halfshaft - LH**.
2. For the inboard constant velocity (CV) joint, remove and discard the boot clamps.
3. Remove the inboard CV joint retaining ring.

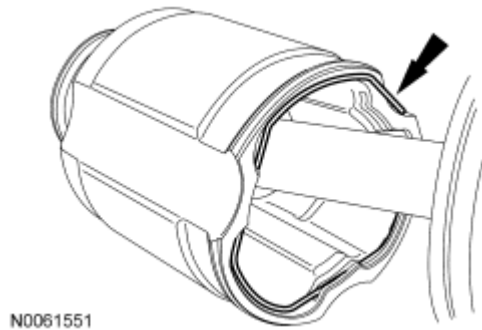


Fig. 37: Locating Retaining Ring
Courtesy of FORD MOTOR CO.

CAUTION: Do not let the roller bearings fall or damage to the component may occur.

4. For the inboard CV joint, carry out the following:
 - Remove and discard the retainer circlip.
 - Slide the boot away from the CV joint.

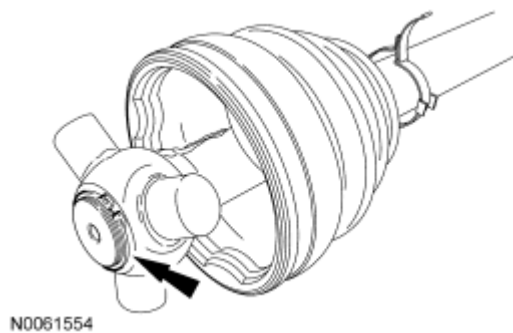


Fig. 38: Identifying CV Joint & Snap Ring On Halfshaft
Courtesy of FORD MOTOR CO.

5. Using a suitable 3-jaw puller, remove the CV joint.

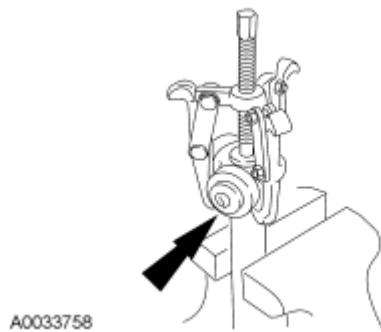


Fig. 39: Identifying CV Joint Using Suitable 3-Jaw Puller
Courtesy of FORD MOTOR CO.

6. Remove and discard the inner CV boot.

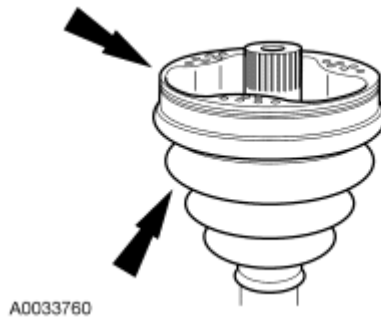


Fig. 40: Locating Inner CV Boot
Courtesy of FORD MOTOR CO.

NOTE: The outboard CV joint is not removable from the halfshaft. The boot must be removed or installed from the inboard CV joint side of the shaft.

7. For the outboard CV joint, carry out the following:
 1. Remove and discard the boot clamps.
 2. Remove and discard the boot.

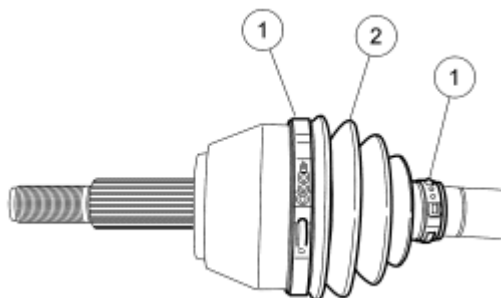


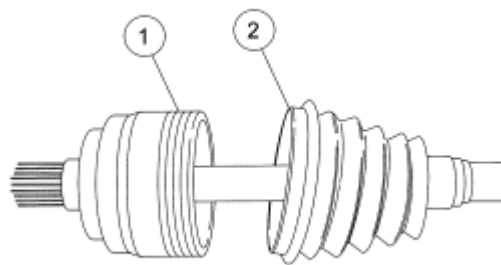
Fig. 41: Identifying Boot Clamps & Boot

Courtesy of FORD MOTOR CO.

ASSEMBLY

NOTE: Do not mix the boot clamps.

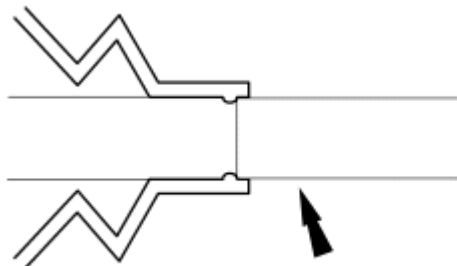
1. Lubricate the outboard CV joint.
 1. Pack the outboard CV joint with grease from the kit.
 - One-third of the grease must be installed in the joint with the remainder placed in the boot.
 2. Spread the remaining grease evenly inside the boot.



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Fig. 42: Identifying Outboard CV Joint & Boot
Courtesy of FORD MOTOR CO.

2. Position the boot so the rib on the boot is located in the groove of the shaft.



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Fig. 43: Locating Boot In Groove Of Shaft
Courtesy of FORD MOTOR CO.

3. Position the outboard halfshaft boot and the outboard boot clamps.
 1. Position the boot.
 2. Position the boot clamps.

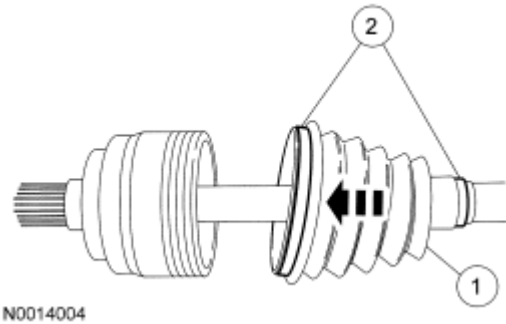


Fig. 44: Positioning Boot Clamps
Courtesy of FORD MOTOR CO.

4. Install the outboard CV boot clamps.

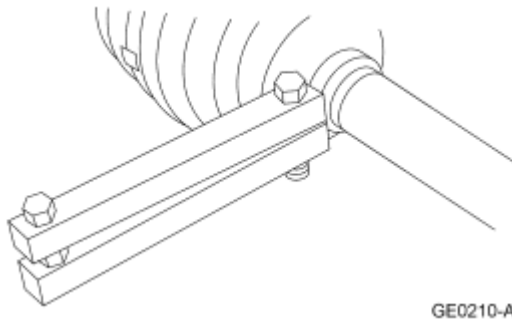


Fig. 45: Identifying Outboard CV Boot Clamp
Courtesy of FORD MOTOR CO.

5. Position the inboard clamp and boot.

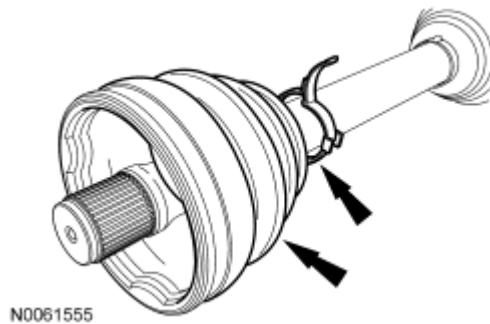


Fig. 46: Identifying Inboard Clamp & Boot
Courtesy of FORD MOTOR CO.

6. Install the CV joint and snap ring on the halfshaft.

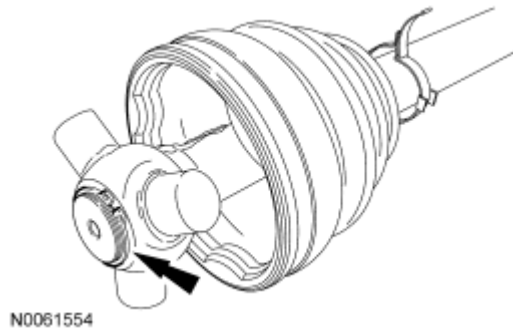


Fig. 47: Identifying CV Joint & Snap Ring On Halfshaft
Courtesy of FORD MOTOR CO.

7. Fill the inboard CV joint housing with grease from the kit.
 - One-half of the grease must be installed in the joint and the remainder placed in the boot.

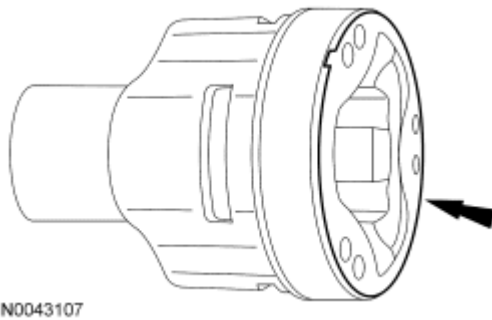


Fig. 48: Locating Inboard CV Joint Housing
Courtesy of FORD MOTOR CO.

CAUTION: Do not let the roller bearings fall or damage to the component may occur.

8. Position the CV housing on the CV joint.

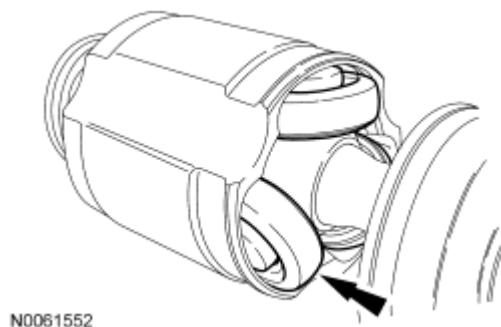


Fig. 49: Locating CV Housing On CV Joint
Courtesy of FORD MOTOR CO.

9. Install the retaining ring.

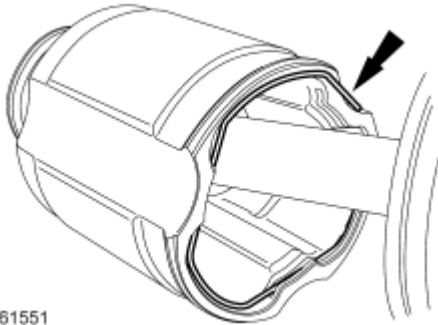


Fig. 50: Locating Retaining Ring
Courtesy of FORD MOTOR CO.

10. Position the inboard halfshaft boot and clamps.
- Position the boot in the housing groove.
 - Position the boot clamps.

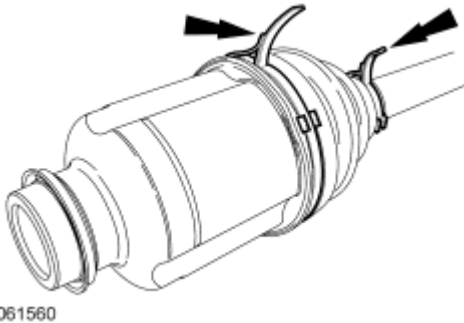
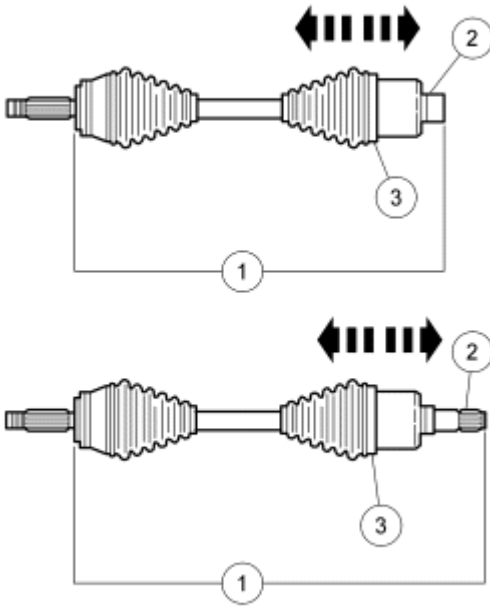


Fig. 51: Locating Boot Clamps & Housing Groove
Courtesy of FORD MOTOR CO.

11. Set the halfshaft assembled length to specifications. For additional information, refer to **SPECIFICATIONS**.
1. Measure the entire assembly length.
 2. Push in or pull out on the inner joint as necessary to adjust the halfshaft assembled length.
 3. Hold the inner joint to prevent the assembled length from changing and insert a small flat-blade screwdriver between the boot and the joint to equalize the pressure.



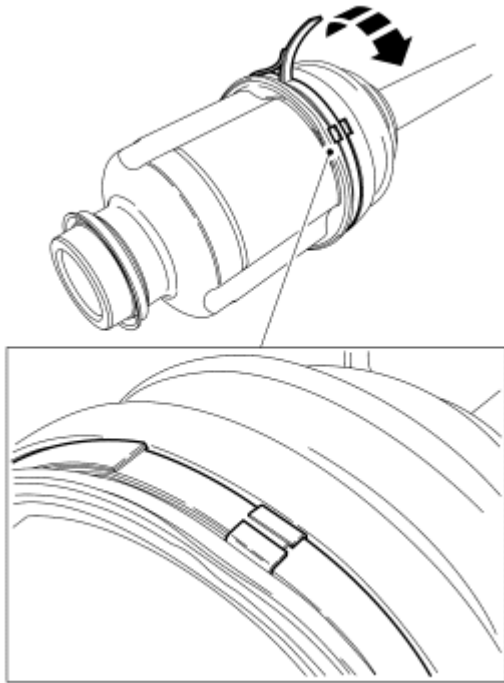
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Fig. 52: Identifying Halfshaft Length Measurements
Courtesy of FORD MOTOR CO.

12. Install the boot clamps.
 - Place the lever arm to the closed position and use a soft-faced hammer to close the tabs.

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N0061556

Fig. 53: Placing Lever Arm To Closed Position & Use A Soft-Faced Hammer To Close Tabs
Courtesy of FORD MOTOR CO.

13. Install the halfshaft assembly. For additional information, refer to **Halfshaft - RH** and **Halfshaft - LH**.